

Rational Episodic Memory: From Rational Action to Agent-Centered Memory in Egocentric Video

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Abstract. Egocentric assistants must continuously process visual streams that can exceed practical storage and computation budgets. Human memory selectively retains goal-relevant information. Drawing on rational-action and Theory-of-Mind accounts, we frame memory admission as inferring which observations are relevant to the wearer’s goals from observable behavior. We present Rational Episodic Memory (REM), which uses behavioral engagement as a functional approximation of agent-centered relevance. Comparing engagement with visual surprise and uniform sampling, we find that engagement performs best on longer recordings, whereas surprise is stronger on shorter recordings, consistent with predictive-processing and event-segmentation accounts of event boundaries.

1 Introduction

Wearable assistants must continuously process egocentric video to answer questions about past experience, such as “Where did I leave my keys?”, but storing every observation exceeds realistic memory budgets. The resulting problem is question-agnostic memory admission: deciding what to retain before the future task or question is known. Humans similarly do not represent visual experiences as an exhaustive record of every retinal input, such as the precise color and texture of every passerby’s shoes. Instead, attention selectively prioritizes information relevant to current goals, making certain visual details more salient, such as noticing those same shoes when considering buying a pair [33].

We draw on rational-action and Theory-of-Mind perspectives to motivate an agent-centered approach to memory admission. Rational-action accounts propose that intelligent agents select actions as means toward goals [16]. Theory of Mind concerns the attribution of unobserved mental states between social agents: an outside agent must infer another agent’s beliefs, intentions, desires, or goals from observable behavior [38]. For example, an observer who sees someone frantically searching their pockets in front of a door can infer the person’s goal is to find their keys. Together, these perspectives define a pipeline from behavior to goal relevance, allowing an observer to infer what an agent is trying to accomplish and treat related experience as potentially important.

A wearable assistant therefore faces the same inference problem. It observes behavior but not the wearer’s internal goals, beliefs, or intentions, so behavior becomes evidence for which ongoing experiences may matter and motivates engagement-based memory admission.

Additionally, surprise is a well-established signal in both cognitive science and vision and serves here as a baseline. Predictive-processing theories describe perception as continually updating expectations about incoming observations [14], while Event Segmentation Theory relates prediction error to event boundaries that shape memory for ongoing experience [36, 54].

2 Related Work

Egocentric-video, embodied- and language-agent systems primarily process completed recordings for summarization [8, 27, 30, 37], retrieval and question answering [7, 11, 41–43], or memory structuring [2, 18, 21, 47]. Online systems organize experience as it unfolds using StartleCam’s physiological triggers [22], SenseCam’s environmental sensors [23], context- and object-centered selection [29, 32], or interaction-based memory structuring in AMEGO [17], which organizes experience around hand-object interactions and locations. Prior work estimates wearer engagement [45], gaze [53], and social interaction [46], but does not use these signals for memory admission.

Computational Theory-of-Mind, inverse reinforcement learning, and human-robot interaction offer precedents for inferring latent goals, states, and intentions from behavior. These precedents include Bayesian generative models [5, 6], learned models such as ToMnet [39], reward-inference approaches [34, 40, 55], and models for assistance, coordination, partner perspectives, and trust [9, 12, 50, 52]. Video-based Theory-of-Mind includes modeling belief dynamics in nonverbal social interaction [13], EgoToM [28] and WAGIBench [49], which benchmark egocentric inference of a wearer’s internal state, and ToMMY [35], which learns to retrieve behaviorally relevant past observations that support inferring an agent’s mental state.

Other precedents include task-relevant observation selection in active perception systems [1, 4] and modern systems such as Clio [31]. Worth Remembering [19] uses predictive surprise as its sole write criterion for question-agnostic online memory admission in robotics.

3 Method

With rational action viewing behavior as organized around goals, we instantiate goal-relevant observation selection with Su and Grauman’s *egocentric engagement* [45]. Their definition treats engagement as goal-directed information gathering, in which the wearer is attracted to an object or scene and interrupts ongoing activity to inspect it more closely. We train a temporal classifier on the Wearable AI benchmark [26, 48], using labels of purposeful information gathering about an object, scene, or task. Producing these annotations requires an annotator to infer intentional, goal-directed behavior from the wearer’s observable behavior, therefore providing a supervised approximation of Theory-of-Mind goal attribution for agent-centered relevance.

V-JEPA 2’s masked latent-feature prediction objective predicts latent representations of masked video, encouraging spatiotemporal structure over pixel

detail [3], making its frozen embeddings a suitable representation for our Random Forest classifier, which operates on 64-frame windows with a 32-frame stride.

Our surprise implementation is an unsupervised baseline inspired by Bayesian-surprise accounts [24, 25]. On frozen V-JEPA 2 embeddings, we compute a dimension-normalized Mahalanobis distance between consecutive windows using diagonal covariance from disjoint development data, then smooth scores before ranking windows.

4 Results

We evaluate 16 Ego4D [20] recordings totaling 42.71 hours, split into eight shorter *event-dense* recordings (1.20-hour average) and eight longer *activity-heterogeneous* recordings (4.14-hour average). The shorter recordings center on one sustained activity or related sequence, whereas the longer recordings span multiple activities and transitions. To proxy useful memory retention, we measure preservation of each Ego4D Natural Language Query (NLQ) evidence interval, which identifies the video context needed to answer the corresponding question [20]. Official intervals cover two long recordings; for the other six, we create 149 narration-derived pairs without access to evaluation scores.

Engagement and surprise are computed incrementally. An offline top- K protocol retains each method’s exact budgeted number of windows, including uniform sampling, to compare ranking under fixed storage independently of deployment-specific thresholding, quota, boundary choices, etc.

Table 1 reports Hit@50, counting an interval when at least half its duration is preserved, and annotated-time recall, the fraction of evidence duration preserved. Engagement performs best at every budget on activity-heterogeneous recordings, whereas Surprise performs best at most budgets on event-dense recordings, suggesting that signal effectiveness depends on temporal structure.

Table 1: Memory retention across temporal regimes.* Entries report Hit@50 / annotated-time recall, in percentages.

Budget	Long recordings			Short recordings		
	Engagement	Surprise	Uniform	Engagement	Surprise	Uniform
30%	42.3 / 44.6	33.0 / 41.5	26.4 / 29.7	15.8 / 15.2	28.8 / 31.6	22.1 / 30.1
25%	33.5 / 38.1	28.8 / 35.9	22.4 / 25.2	12.0 / 13.0	24.0 / 27.8	17.6 / 25.0
20%	30.7 / 32.6	20.5 / 29.6	18.6 / 20.2	7.9 / 8.8	17.5 / 23.3	14.5 / 19.9
15%	25.6 / 27.5	13.5 / 21.5	13.0 / 15.0	7.2 / 7.2	11.0 / 15.9	10.9 / 15.0
10%	16.7 / 18.1	9.3 / 14.0	9.0 / 10.0	3.8 / 5.2	6.2 / 10.9	7.3 / 10.0
5%	10.7 / 11.1	4.2 / 7.3	4.2 / 4.8	2.7 / 3.2	2.4 / 5.4	3.5 / 4.9

* An annotation-source sensitivity analysis found no evidence that researcher-created intervals biased the result: Engagement outperformed Surprise on Hit@50 for both sources, with less consistent recall differences reflecting video-level variation. Additionally, an uncertainty analysis was directionally consistent, but small-sample video variation produced wide intervals that often crossed zero.

5 Discussion

The reversal suggests distinct roles: Surprise highlights prediction error and event boundaries [14, 36, 44, 54], whereas engagement approximates agent-centered relevance through purposeful goal pursuit [16, 38]. Engagement is more discriminative when goal-directed behavior is sparse but may saturate in event-dense recordings. Per-recording means support this interpretation: predicted engagement is higher and more tightly clustered in the short subset (0.67–0.77; mean 0.72) than in the long subset (0.52–0.70; mean 0.60), consistent with saturation when one activity dominates. Event segmentation may help distinguish local changes; engagement may identify sparse goal-directed behavior.

To separate duration from activity structure, we analyze a 5.2-hour recording dominated by one sustained task, *welding* (mean engagement = 0.678). It is excluded from the primary aggregate because it is structurally distinct from the activity-heterogeneous subset. Table 2 shows Surprise higher on Hit@50 at moderate and high budgets, while Engagement is stronger at tighter budgets and retains more annotated duration. The case suggests that temporal structure shapes signal utility, with Surprise more discriminative when engagement is elevated throughout.

Table 2: Retention on the long, task-saturated recording. In percentages.

Budget	Hit@50		Annotated-time recall	
	Engagement	Surprise	Engagement	Surprise
30%	26.3	61.4	42.1	52.2
25%	24.6	42.1	41.5	40.8
20%	17.5	29.8	34.8	30.1
15%	15.8	19.3	29.4	22.2
10%	14.0	3.5	22.8	10.8
5%	7.0	1.8	10.6	0.3

Limitations include the small 16-recording sample, offline top- K /NLQ evaluation, researcher-created and single-annotator labels, uniform-only comparison, no cross-user testing, and coupled duration and activity composition.

Here, surprise measures local changes in frozen V-JEPA 2 features. Friston’s predictive-processing account posits an internal generative model that predicts sensory input from latent causes, with mismatches producing prediction error linked to event boundaries [15, 36, 54]. EgoSelf and EgoAgent provide structured and neural precedents for user-specific interaction and world-state prediction [10, 51]. A future implementation could adapt the V-JEPA 2 predictor to estimate wearer-specific expectations for recurring environments, objects, routines, and interactions rather than using only frozen encoder features [3].

We presented a cognitively inspired pipeline for question-agnostic egocentric memory. Rational action defines relevance, Theory of Mind frames the system as inferring aspects of the wearer’s internal state, and behavioral engagement provides a functional approximation for memory admission. Surprise provides a complementary signal grounded in predictive-processing and event-segmentation accounts.

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